

Beicheng Xu

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Automated Research / AutoML · AI for Systems · Agents

Education

Peking University School of Computer Science Ph.D. student in Computer Science and Technology Advisor: Prof. Bin Cui (IEEE Fellow)	Sep. 2023 – Present
Wuhan University School of Computer Science Bachelor's degree in Computer Science and Technology GPA: 3.99/4.00 Rank: 1/319	Sep. 2019 – Jun. 2023

Publications

Area I: AutoResearch / AutoML

CoFEH: LLM-driven Feature Engineering Empowered by Collaborative Bayesian Hyperparameter Optimization B Xu, K Ding, W Liu, Y Lu, B Cui <i>ACM SIGKDD Conference on Knowledge Discovery and Data Mining</i>	KDD 2026 Oral
Tree-Structured Synergy of Large Language Models and Bayesian Optimization for Efficient CASH B Xu, W Qian, L Tung, Y Lu, B Cui <i>Neural Information Processing Systems Annual Conference</i>	NeurIPS 2026 Poster
PSEO: Optimizing Post-hoc Stacking Ensemble Through Hyperparameter Tuning B Xu, W Liu, K Ding, Y Lu, B Cui <i>AAAI Conference on Artificial Intelligence</i>	AAAI 2026 Oral
HyperRPS: An Efficient Multi-Fidelity Hyper-Parameter Tuning Framework for Rock Particle Segmentation Y Ren*, B Xu*, H Jiang, X Liu, P Zhang, Q Liang, Y Shen, B Cui (co-first author) <i>Tsinghua Science and Technology</i>	TST 2026 (JCR Q1)

Area II: AI for Systems

MFTune: An Efficient Multi-fidelity Framework for Spark SQL Configuration Tuning B Xu, L Tung, Y Wang, Y Lu, B Cui <i>International Conference on Very Large Databases</i>	VLDB 2026
A-Tune-Online: Efficient and QoS-Aware Online Configuration Tuning for Dynamic Workloads Y Shen*, B Xu*, Y Lu, D Chen, H Jiang, Z Xie, S Fu, N Zhang, Y Ren, N Jia, X Hu, B Cui (co-first author) <i>International Conference on Data Engineering</i>	ICDE 2025

Projects & Internships

AutoSci: Automated Research System Supports end-to-end research automation, covering literature review, ideation, experimentation, and writing, including AI for AI: feature engineering, algorithm selection, hyperparameter tuning, and model ensembling.	Project Lead	GitHub 1.7k stars
Openbox: General-Purpose Black-Box Optimization Supports single-objective and multi-objective black-box optimization.	Project Lead	GitHub 431 stars
Tencent TEG Intelligent Algorithms Team: AI for System Optimize the Hunyuan AI Data pipeline.	Industry Collab. (Intern)	Jul. 2026 – Present
Huawei HiSilicon: General-Purpose Offline System Tuning Offline configuration tuning for DB, Spark, Ceph, and HPC with high evaluation costs (VLDB 2026).	Industry Collab. (Intern)	Nov. 2024 – Nov. 2025
Huawei 2012 Labs: General-Purpose Online System Tuning Online configuration tuning for DB, Spark, Kafka, Redis, and Unix under dynamic workloads (ICDE 2025).	Industry Collab. (Intern)	May 2023 – May 2024
PetroChina: Rock Particle Segmentation Auto-Tuning Multi-fidelity hyperparameter tuning for efficient, accurate rock particle segmentation (TST 2026).	Industry Collab. (Intern)	Jul. 2022 – Jul. 2024

Honors & Awards

National Scholarship (twice)	2020–2021; 2021–2022
BYD Scholarship, Peking University	2025–2026
Merit Student (4 times)	2019–2022; 2025–2026
Ranked 1st in Comprehensive Evaluation (3 times)	2020–2021; 2021–2022; 2025–2026
First-Class Scholarship, Wuhan University (3 consecutive years)	2019–2022
CCF Outstanding Undergraduate Award (102 recipients nationwide)	2022
Outstanding Graduate; Outstanding Bachelor's Thesis, Wuhan University	